

AI Features

Enterprise Learning Platform

Documentation of AI-Powered Capabilities

1. Overview

Enterprise Learning Platform is a Learning Management System that uses AI to make learning more personal and to save mentors time on repetitive work. This document explains, in plain terms, what each AI feature does, how it works behind the scenes, and where in the platform it is used. It is meant as a reference for anyone reviewing this repository who wants to understand the AI layer without reading through the source code.

The platform has four AI-powered features, and all of them are built on top of a single, shared AI engine rather than four separate integrations. This keeps the system consistent and easy to maintain — if the underlying AI model is ever swapped out, only one part of the codebase needs to change, and every feature that depends on it keeps working exactly as before.

The four features are:

- AI Chat Assistant — a conversational tutor for students
- AI Quiz Generator — automatic quiz creation for mentors
- Personalized Learning Roadmap — a monthly study plan built from a student's real activity on the platform
- Skill Gap Analysis — a comparison of a student's current skills against what a target job role actually requires

Each feature is described in detail below, followed by an explanation of how the shared AI engine works and how the whole system is configured.

2. AI Chat Assistant

What it does

This is a general-purpose AI tutor available to students from their dashboard. Students can ask questions on almost any subject — programming, mathematics, science, career advice, or general knowledge — and get a clear, well-formatted answer. At the end of every answer, the assistant also suggests a few natural follow-up questions, so the conversation can continue smoothly instead of dead-ending.

How it works

When a student sends a message, the system first checks whether there is an existing conversation to continue. If so, it pulls in the last several exchanges from that conversation so the AI has context and does not lose track of what was discussed earlier. The message, along with this recent history, is sent to the AI engine, which returns a formatted response. That response is shown to the student and is also saved, together with the original question, so it becomes part of the conversation history for next time.

Every conversation is tracked using a unique identifier, which allows a student to have multiple separate chat threads without them mixing together. The amount of history the assistant remembers is configurable, so it can be tuned for performance or cost without changing how the feature behaves.

Where it's used

This feature lives on the student dashboard as an always-available chat panel, functioning as a personal tutor a student can turn to at any point during their learning journey.

3. AI Quiz Generator

What it does

Instead of manually writing quiz questions, a mentor can select a lesson or topic and have the AI generate a complete quiz automatically — including multiple-choice questions, four answer options each, the correct answer, a short explanation, and a difficulty level. The mentor reviews the generated quiz and can publish it with a single click, or make edits first.

How it works

When a mentor requests a quiz, the system gathers relevant context about the lesson so the questions are grounded in real course material rather than being generic. This context, along with the desired number of questions and difficulty level, is sent to the AI engine, which returns a structured set of questions. The system validates this output before showing it to the mentor, to make sure every question actually has the right number of options and a valid correct answer.

If the AI service is temporarily unavailable, the quiz generator does not simply fail — it falls back to a local question bank built around common topics, such as frontend development, backend development, databases, and machine learning. This ensures mentors are never blocked from creating a quiz, even during an AI outage.

Time limits for the quiz are calculated automatically based on the number of questions, and correct answers are always hidden from students once the quiz is published, to preserve quiz integrity.

Where it's used

This feature is part of the mentor dashboard, specifically within the lesson and quiz management area, where mentors build and publish assessments for their courses.

4. Personalized Learning Roadmap

What it does

This is the most data-driven feature in the platform. Rather than giving every student the same generic study plan, the system builds a roadmap based on that specific student's real history and performance — their profile, courses, quiz scores, assignment grades, prior skill gaps, and even the topics they have previously asked the AI chat assistant about.

The result is a month-by-month plan, typically covering four to six months, with weekly milestones, specific skills to focus on, recommended projects, and interview preparation guidance — all tailored to help the student progress toward their stated career goal.

How it works

When a student requests a roadmap, the system pulls together information from six different sources: the student's profile and stated career goal, their enrolled and completed courses with progress percentages, results from any previous skill gap analysis, their quiz performance history, their assignment and project grades, and recent topics from their AI chat history.

All of this is combined into a single, detailed picture of the student and sent to the AI engine with clear instructions: every stage of the roadmap must address the student's weaker areas, must include practical projects and interview preparation, and must be grounded in the student's actual data rather than generic advice. A student can also optionally specify a target topic, such as Full Stack Development, in which case the entire roadmap is focused around that goal.

The generated roadmap is saved to the student's profile so it persists across sessions and is displayed as a series of stage cards the student can track their progress against.

Where it's used

This feature appears on the student dashboard as a dedicated Learning Roadmap section, giving students a clear, personalized path forward instead of having to figure out what to study next on their own.

5. Skill Gap Analysis

What it does

A student enters a target job role — for example, Full Stack Developer or Data Scientist — and the AI compares their current skill set against what that role realistically requires in the industry. It identifies exactly which skills are missing, which ones matter most to prioritize, and recommends specific courses already available on the platform to close those gaps.

How it works

The system first builds a picture of the student's current skills by combining several sources: skills listed directly on their profile, their stated interests, the courses they have enrolled in (which imply certain skills), and topics from quizzes they have already passed with a good score. This combined, de-duplicated skill list — along with the target role and a list of currently available courses — is sent to the AI engine.

The AI returns an analysis identifying the student's current skills, the skills required for the target role, the specific gaps between the two, and which of those gaps should be prioritized first. It also suggests general recommendations and relevant keywords.

The system then takes those keywords and automatically searches the live course catalogue for matches, so the student is not just told what they are missing — they are pointed directly to real courses on the platform that can help them close the gap. The top matching courses are returned alongside the analysis.

Where it's used

This feature is available on the student dashboard, typically as part of a career-planning or profile section, and is often used together with the Learning Roadmap feature since the roadmap can build on a prior skill gap analysis.

6. How the Shared AI Engine Works

All four features described above rely on the same underlying AI engine rather than each having its own separate integration. This engine is responsible for actually communicating with the AI model and returning a response, regardless of which feature is asking.

Reliability through fallback

To keep the platform functional even if one AI provider has an outage, the engine is designed to try multiple models in sequence. It starts with the primary configured model, and if that fails or returns no result, it automatically tries the next one in line, continuing through a short list of alternative free and paid models until one succeeds. Only if every option in the chain fails does the system report an error — and even then, certain features, such as the Quiz Generator, have their own local fallback so the user experience is never fully interrupted.

Flexible provider support

The engine can operate in two modes depending on how it is configured: it can talk directly to a primary AI model, or it can route requests through an alternative gateway that unlocks the multi-model fallback chain described above. Which mode is active is determined automatically based on the type of API key configured, so switching between the two requires no code changes — only a configuration update.

Why this matters for the rest of the codebase

Because every feature depends only on a general "ask the AI" interface — and not on the specific details of which model or provider is being used — new AI capabilities can be added to the platform in the future, or the underlying model can be upgraded or replaced, without needing to touch the logic of the Chat Assistant, Quiz Generator, Learning Roadmap, or Skill Gap Analysis features individually.

7. Configuration Summary

The AI engine is controlled through a small set of configuration values:

- Provider selection — determined automatically by the format of the configured API key.
- Model choice — which specific model is used as the primary option when running through the fallback gateway.

- Conversation memory length — how many past messages the Chat Assistant considers for context, which can be tuned for performance or cost.

These settings are kept outside the application code so the AI behavior can be adjusted without redeploying the platform.

8. Summary

Feature	Used By	Purpose	Has Fallback
AI Chat Assistant	Students	General-purpose conversational tutor	No
AI Quiz Generator	Mentors	Automatic quiz creation from lesson content	Yes — local question bank
Personalized Learning Roadmap	Students	Monthly study plan based on real activity	No
Skill Gap Analysis	Students	Compares skills to a target role, suggests courses	No

Together, these four features form a cohesive AI layer that supports students throughout their learning journey — from day-to-day questions, to assessment, to long-term planning — while giving mentors AI-assisted tools to build course content more efficiently.